

Project Documentation: Blinkit Sales & Operations Analytics

Project Overview A comprehensive analysis of 8,523 transaction records to optimize sales performance and outlet efficiency for Blinkit. The project identifies key revenue drivers across product categories, outlet locations, and store formats.

1. Technical Framework & Data Cleaning

The raw data underwent a rigorous cleaning process to ensure 100% accuracy in reporting:

- **Standardization:** Unified `Item_Fat_Content` labels into "Low Fat" and "Regular."
- **Imputation:** Missing `Item_Weight` and `Outlet_Size` values were filled using Median/Mode based on item types and location tiers.
- **Zero-Value Correction:** Impossible 0 values in `Item_Visibility` were replaced with category medians.

2. Feature Engineering (New Metrics)

Two strategic columns were engineered to provide deeper business insights:

- **Outlet Age:** Calculated as $2023 - \text{Establishment_Year}$ to measure maturity-linked growth.
- **Visibility Efficiency:** Calculated as $\text{Total_Sales} / \text{Item_Visibility}$. This identifies products that generate high revenue relative to their display space (Sales ROI).

3. Key Statistics & Business Insights

- **Total Revenue:** ₹1.20 Million with an average customer rating of **3.96/5**.
- **Top Performers:** **Fruits & Vegetables** and **Snack Foods** contribute over ₹350k combined.
- **The "Sweet Spot":** **Medium-sized outlets in Tier 3 cities** are the most profitable, outperforming Tier 1 locations by nearly 40%.
- **Efficiency Leader:** The **"Meat"** category shows the highest **Visibility Efficiency**, proving it is a "natural seller" that requires less promotion to drive high sales.

4. Dashboard Architecture

The analysis is presented across three specialized dashboard views:

1. **Executive Overview:** High-level KPIs (Revenue, Avg Sales, Ratings) for senior leadership.
2. **Product Intelligence:** Deep dive into category performance and fat-content trends.
3. **Outlet Efficiency:** Analysis of store size, location tiers, and historical growth patterns.

5. Strategic Recommendations

- **Expansion:** Prioritize "Medium" store formats in Tier 3 regions for the highest ROI.
- **Marketing:** Reallocate digital shelf space from low-efficiency categories (like Seafood) to high-efficiency drivers (like Meat) to maximize revenue per view.
- **Optimization:** Audit underperforming stores established in 2011 to modernize operations to 2018 standards.

Tools Used: Google Sheet (Data Cleaning, Pivot Tables, Dashboard)